

The Phantom Diaries: The Spectral Train

Official Game Document

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Elevator Pitch

'The Phantom Diaries: The Spectral Train' is a story-driven, horror-puzzle escape room experience set aboard a haunted 1920's steam train where every solved puzzle pulls you closer to the truth – or the demon...

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Game Concept

Game Summary

The Phantom Diaries: The Spectral Train originated as a small, escape room-inspired passion project for an end-of-year college assignment. After receiving high praise from tutors, it has progressed into a fully realized indie title, built on advanced game development skills refined over many years of education.

Set in 1923 during the golden age of American railways, players control police detective **Thaddeus Mercer** on a steam train journey home. Things soon take a dark turn when Mercer discovers the train is possessed by an evil demon who intends to recreate a fatal derailment from 20 years prior.

Players must navigate each locked train carriage and search for clues to solve puzzles that hold the keys needed to advance. Ghosts can be interacted with for aid in these puzzles and/or to give information about the world, while the demon remains a constant threat, waiting for opportunities to strike.

Players must utilize their Inventory Menu, where they can review collected items in the Information Menu or connect them through the Investigation Menu to piece together puzzles – reflecting Mercer’s analytical mindset as a detective, placing logical deduction at the centre of the gameplay experience.

The game is loosely inspired by the real-world event ‘*The Wreck of the Old ‘97*’.

Genre & Target Audience

The Phantom Diaries: The Spectral Train is a first-person, horror-puzzle game designed for players who enjoy atmospheric, story-driven gameplay experiences with tense horror delivered through environmental storytelling and jumpscares. The game is also suitable for puzzle enthusiasts who prefer minimal handholding.

Recommended age: **PEGI 16** – Strong fear, strong horror, gambling themes.

Target Platform

The game targets Windows PC only. Mobile support is not feasible, while console support is currently beyond scope.

Unique Selling Points

- **Immersive Escape Room Horror:** Explore a haunted 1920's steam train filled with intricate puzzles and terrifying horror elements.
- **Evolving Narrative:** Slowly unravel an engaging story through puzzles and exploration.
- **Historical Inspiration:** Loosely based on the real-world 1903 event *The Wreck of the Old '97* to offer authenticity.
- **Niche Genre:** The game's hybrid genre targets a niche genre within the gaming market, giving it a distinct position.

Story & Setting

Premise

In September 1923, protagonist **Thaddeus Mercer**, a 27-year-old police detective, boards *The Old Royal Express Line* steam train to travel from Monroe, Virginia to his home in Spencer, North Carolina.

During the journey near Danville, Virginia, Mercer encounters the ghost of the train's former conductor and discovers the train is not only haunted, but also possessed by an evil demon named **Mammonach**, who aims to repeat the deadly derailment of the train from 20 years ago. Mercer must use his expertise to solve a variety of puzzles enroute to the Controls Carriage to prevent history repeating itself. The further he advances, the more the story unfolds to tell a gripping and unpredictable tale of greed, persuasion, and evil.

The story is loosely inspired by the real-world event *The Wreck of the Old '97*, where a train derailed near Danville, Virginia in 1903 due to excessive speeds.

World & Environment

The game takes place aboard a classy steam train with heavy steampunk, Victorian, and Western aesthetics to help create a luxurious interior that contrasts with supernatural horror. Outdoor views of the train are limited to looping mountain and train track scenery to maintain immersion within the setting. The environment has been created to match real-life American railway aesthetics from the early 1900's – the setting and time period of the game.



Figure 1: The Reception Carriage of the train, displaying luxurious decorations inspired by steampunk, Victorian, and Western aesthetics.



Figure 2: The Seating Carriage of the train, further displaying more luxurious interior design inspired by steampunk, Victorian, and Western themes.

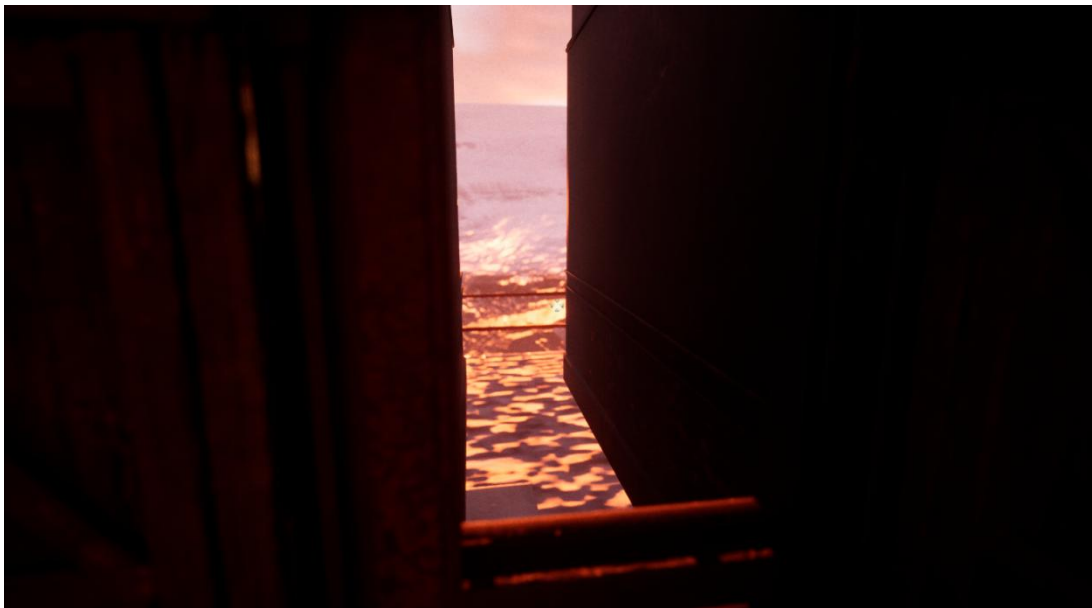


Figure 3: A small section of exterior space between two train carriages, showcasing the outside mountain range and train tracks.

Characters

Thaddeus Mercer (Protagonist)

A 27-year-old police detective on his way home to Spencer, North Carolina onboard *The Old Royal Express Line* steam train. Observant, smart, and analytical, Mercer uses his skills as a detective to piece together clues using the Investigation Menu to complete puzzles within the game. Equipped with a police standard flashlight.

Mammonach the Demon (Antagonist)

A once greedy and selfish man named Marcellus Mayweather who owned *The Old Royal Express Line*, Mammonach was born from Marcellus' greed. The malevolent force driving the possession of the train, the culprit of the train's crash 20 years prior, and the end goal for players to defeat.



Figure 4: Mammonach the Demon.

Ghosts (Side Characters)

The victims of the fatal crash from 20 years ago, bound to ride the train for eternity. Each with unique quirks and histories, they offer quests, clues, information, and time bonuses that shape Thaddeus' journey. Each ghost reflects a fragment of the train's past, from a duty-bound conductor to a scheming gambler to a bellboy forever tidying luggage, and more.



Figure 5: The ghost of the train's conductor, Thomas Hanksworth.



Figure 6: The ghost of a young woman passenger named Margot 'Maggie' Sinclair.

The Watcher (Narrative Character)

A mysterious figure influencing events from the shadows. Never directly encountered, but ever-present through collectibles and narrative breadcrumbs.

Narrative Flow

The story unfolds through ghost interactions and collectibles, allowing players to uncover as much – or as little – of the story as they choose. This supports both minimal and lore-rich playthroughs alike, encouraging exploration without forcing exposition.

Although a handful of story developments will be discovered mandatorily, more than half of the full picture is discoverable voluntarily via lore collectibles, ghost interactions, and general exploration, giving players complete freedom over their experience of the game's narrative.

Additionally, the discoverable story pieces become increasingly detailed the further players advance, ensuring a smooth and steady narrative progression early before ramping up the depth later on.

Gameplay Overview

Core Mechanics

The games' small environment and detailed story approach empathizes exploration and puzzle solving over advanced mechanics. Players utilize four primary actions:

- **Movement:** To traverse the level.
- **Interacting:** To interact with interactable items and characters.
- **Flashlight Toggle:** To turn the flashlight on and off.
- **Pause:** To pause the game and showcase the Pause Menu.

Objectives

The main objective is to reach the Controls Carriage to stop the train from speeding into a crash before the timer runs out. Players have 30 minutes post-tutorial to accomplish this, which can alter through ghost interactions.

To advance to the Controls Carriage, players must solve the other carriages' puzzles in order to find the keys to advance. These puzzles act as the side objectives of the game and are mandatory for completion. Ghost interactions and collectibles are not mandatory but can offer important information to the whereabouts of the keys, along with additional story developments and other useful information.

Rules & Systems

The game features a timer system inspired by real-life escape rooms which counts down as soon as the tutorial is completed at the beginning of the game. The timer can be increased or decreased through ghost interactions, and if the timer runs out, players can either restart from their latest autosave or quit to the Main Menu.

Each carriage contains an autosave checkpoint at its entrance which can be used for resetting the game following the timer running out or upon death. Players can also manually trigger a reset through the Pause Menu if they wish.

Players has full freedom in their approach to the game in regard to their interactions and exploration but are required to complete a handful of puzzles within the train carriages to advance to the finale of the game, where they can beat the demon and stop the evolving madness.

Finally, as previously mentioned, there is a tutorial to complete at the beginning of the game before the timer begins which aims to increase player familiarity with the mechanics and gameplay systems.

Progression

Progression gradually ramps up both horror and narrative intensity. Earlier carriages feature lighter scares and surface-level story developments, while later ones deliver heavier plot reveals and scarier horror experiences. This approach keeps players engaged by making them earn a more challenging, scary, yet story-rewarding experience the more they advance and uncover.

The demon encounters are scripted for specific moments and will appear more often as players advance further. This allows players to gain familiarity with the game's controls and how to combat the demon before throwing multiple encounters their way – while also feeling natural to the story (where the demon would narratively appear more often as players get closer to completing their goal).

Level Design

Level Layout

The level is split into six separate train carriages, each representing a unique theme and puzzle designed to its function. These are (from first encountered to last): Reception Carriage, Bar Carriage, Dining Carriage, Seating Carriage, Storage Carriage, and Controls Carriage.

To keep the game looking and feeling fresh (and to prevent repetitive experiences) each carriage has a distinct visual identity, with its design and furnishings reflecting its function. Each carriage contains a unique puzzle for players to solve to find the key for the next carriage, which matches the function of the carriage they are in.

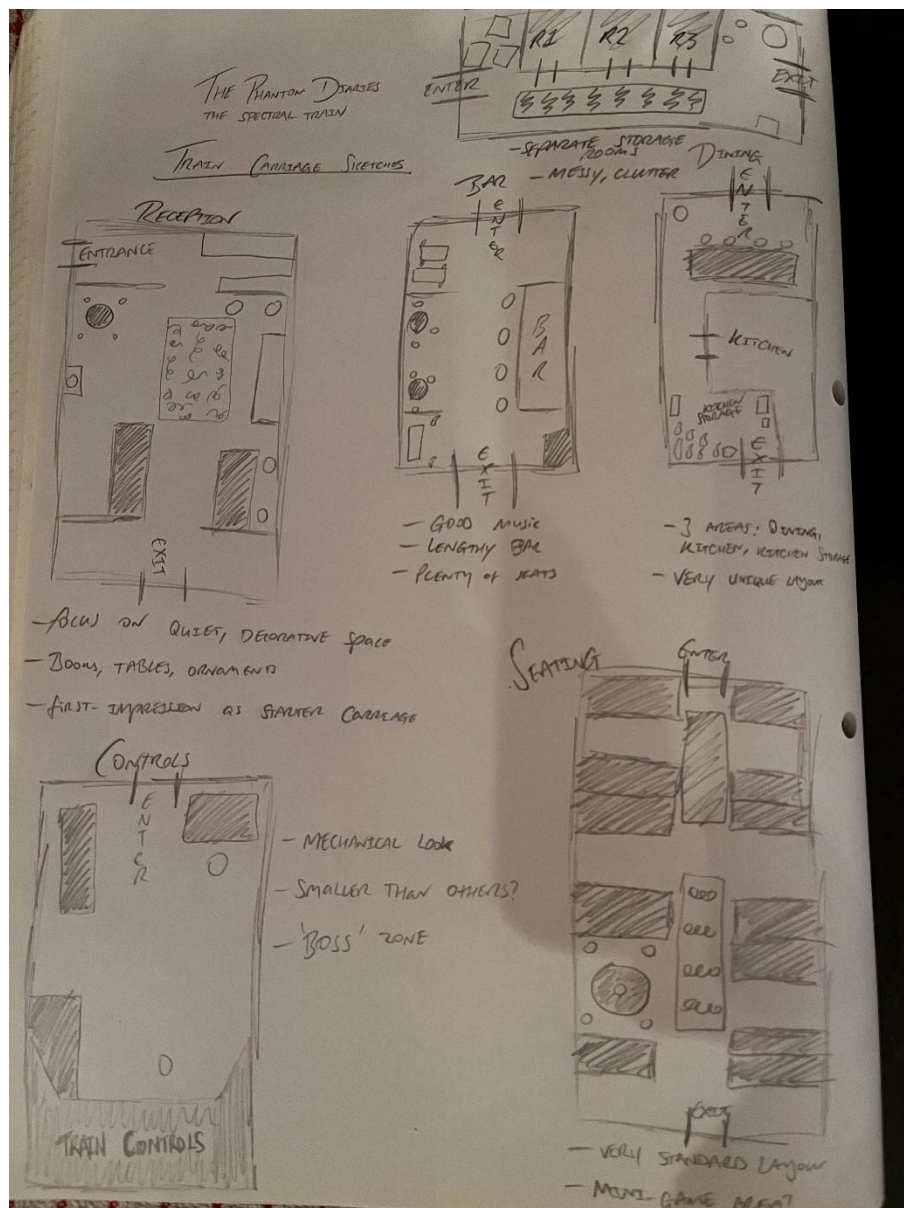


Figure 7: Hand-drawn sketches for each carriage layout, created during the game's pre-production phase.

Challenges & Puzzles

Demon encounters serve as quick-time events where players must shine their flashlight at the demon before they are attacked; failure results in death and reset. Additionally, each carriage features its own function-themed puzzle to match the carriage's setting – these puzzles are explained in more detail in the table below.

Carriage	Puzzle Type	Mechanic	Theme
Reception Carriage	Key in book on specific page number	Search collectibles for specific page number	Reading
Bar Carriage	Key in 3-digit-lock chest	Clue puzzle related to alcoholic drinks from the Bar	Drinks
Dining Carriage	Key in 3-digit-lock chest	Observation puzzle related to meals from a menu	Food
Seating Carriage	Mini game	Liar's Dice mini game versus a ghost	Ghost interaction
Storage Carriage	Hidden in plain sight	Environmental exploration	Trick/false puzzle
Controls Carriage	Multi-carriage runback sequence	Multiple demon encounters	Climax/final battle

Rewards & Progression

The game delivers a steady progression of rewards by gradually increasing the frequency of scares and demon encounters and producing more flashlight batteries and deeper story revelations the further players advance. The difficulty will escalate both narratively and mechanically, rewarding persistence.

Art & Visuals

Art Direction

The game uses a realistic art direction to maximize immersion and elevate the horror atmosphere through believable environments and lighting – a method used by popular horror game franchises such as ‘*Outlast*’, ‘*Silent Hill*’, and ‘*Resident Evil*’ – all of which served as inspirations for this game’s art direction. Additionally, due to the game’s frightening nature with intentions of immersion and thrills, all other art directions aside from realism did not suit the intended immersive horror experience.



Figure 8: Environment from *Resident Evil Village*, showcasing the realistic art style.

Screen Rant (2021). Available at: <https://screenrant.com/re-village-graphics-settings-best-resident-evil-game/>
(Accessed 17th March 2026)



Figure 9: Environment from *Outlast*, further showcasing the immersion realistic art styles can deliver for horror experiences.

GamingBolt (2014). Available at: <https://gamingbolt.com/outlast-whistleblower-dlc-arriving-in-april-for-ps4-and-pc>
(Accessed 17th March 2026)

Characters & Environment

The player character is not visually represented, as is common with most first-person games where embodiment is not a core focus. The demon is represented by a realistic monster design that seamlessly fits into the world. The ghosts are represented by semi-transparent, blue-tinted models with minimal facial detail but high body detail showcasing their apparel and working status. The demon and ghosts can be seen in pictures further up the document.

The environment is a high-fidelity interior of a classy and luxurious steam train with era-authentic furnishings and strict personality designs for each carriage to match their function, while the outside area surrounding the train features looping mountain scenery to match the setting of the game, with train tracks underneath to further increase immersion.

The environment itself has been made larger in-game to cater to a better playing experience with more room to traverse rather than a faithful one-to-one dimension scale recreation of train interiors from the early 20th century, where movement would be extremely limited and would therefore impact gameplay significantly.



Figure 10: Interior from a real train in the early 20th century. Rare Historical Photos (2025). Available at: <https://rarehistoricalphotos.com/vintage-train-travel-photos-early-20th-century/> (Accessed 18th March 2026)



Figure 11: More train interior from the early 20th century.
 Rare Historical Photos (2025). Available at: <https://rarehistoricalphotos.com/vintage-train-travel-photos-early-20th-century/> (Accessed 18th March 2026)



Figure 12: Additional interior of a real train from the early 20th century.
 Rare Historical Photos (2025). Available at: <https://rarehistoricalphotos.com/vintage-train-travel-photos-early-20th-century/> (Accessed 18th March 2026)



Figure 13: Even more real train interior from the early 20th century.
Rare Historical Photos (2025). Available at: <https://rarehistoricalphotos.com/vintage-train-travel-photos-early-20th-century/> (Accessed 18th March 2026)

HUD & UI Elements

The HUD is minimal in design, showcasing only essential elements to keep the experience fully immersive, such as:

- **Aiming Cursor (Centre):** Helps players aim their interaction line to wherever they want.
- **Interaction Prompt (Bottom-Centre):** Displays prompts for players to interact with interactable objects and characters.
- **Notification Pop-Up (Top-Left):** Showcases important notifications such as autosaves, locked doors and chests, or when the player cannot use a flashlight battery due to having full capacity.
- **Flashlight Battery Indicator (Bottom-Left):** Reveals a flashlight icon to the player whenever their flashlight is on, as well as a countdown timer of the current battery value for players to keep track of.



Figure 14: The HUD.

The Main Menu appears transparent over the background of the game's opening cutscene, creating a seamless transition between menu and gameplay, featuring:

- **Game Title:** Big and bold.
- **Play and Load Buttons:** For starting a new game or loading a game from save.
- **Options Button:** For viewing the controls scheme or editing a variety of options, such as resolution, audio volume, and brightness.
- **Quit Button:** For quitting the game.



Figure 15: The Main Menu.

The Pause Menu overlays most of the game screen with a canvas in the middle to display information and buttons on the side of the canvas for switching between the menu elements. The Pause Menu features:

- **Home Page:** Displays the game title, the current carriage the player is in, and the remaining game time.
- **Inventory Page:** Displays all the collected items of the player.
- **Investigation Page:** Can only be entered by clicking on a *Clue Item* from the Inventory Page. Showcases the currently selected clue and any other related clues or missing clues.
- **Information Page:** Can only be entered by clicking on a *Lore Item* from the Inventory Page. Showcases the detailed information of the item.
- **Options Page:** Same function as the Main Menu.
- **Reload Button:** Reloads the game from the player's latest autosave.
- **Quit Button:** Quits to the Main Menu.



Figure 16: The in-game Pause Menu.

The HUD and menus use Western-looking, early 20th century fonts with a burgundy and cream-yellow colour scheme to match the train's identity and the setting of the game. Authentic handwritten and accessibility typewriter fonts enhance immersion within the Investigation and Information Pages when reading item information and clues.



Figure 17: The Options Menu, accessible from the Pause Menu.



Figure 18: The authentic font of the game for items with readable information.



Figure 19: The accessibility font for easier reading.

Animations

Player

The player uses a camera shake for idle and walking movements to simulate first-person animations but does not have any body animations due to not having a body model.

Demon

The demon has screaming and chasing animations that are used for its mechanics where, after spawning, the demon screams before chasing the player to offer a small reaction buffer to the scenario.

Ghosts

The ghosts remain fixed in place throughout the game, but their bodies animate with a glow effect to showcase their ghostly nature.

Environment

Environmental objects such as lanterns, chandeliers, doors, books, and chests feature dynamic animations to bring the world to life, while the exterior mountains and train tracks animate to simulate the movement of the train, giving the illusion that the train itself is moving.

Cinematics

The game includes opening and ending cutscenes to tell the story, along with dedicated sequences for death and timer expiration, should these occur.

Audio

Music

Authentic 1920's phonograph music is used from source-free libraries to further the immersion of the experience. Having these play from phonographs dotted throughout the level not only helps build atmosphere but solidifies the setting of the game.

Furthermore, era-specific ambient score, such as guitar and violin pieces, is employed for cutscenes and menus to build cozy or tense moods when required.



Figure 20: A phonograph from the game, which plays music.

Sound Effects

Interactions within the game use context-sensitive sound effects, such as squeaks for the opening of doors and chests, paper rustles for the opening of books or collection of pamphlets, lock shakes for locked containers, flicks for light switches, key jangles for key collections, and more.

Thematic notification tones are used to fit the music style and to alert players to any incoming notifications. The demon utilizes distinct and recognizable spawn and chase cues to help players gain familiarity with its spawning and mechanics.

Finally, appropriate and professional UI sounds, such as button clicks, are used throughout the menus to create a reactive and enjoyable user experience.

Controls & User Interface

Controls

- **Movement (WASD):** Allows the player to traverse the game world.
- **Interacting (E):** Allows the player to interact with collectibles, in-game moveable objects such as doors and chests, and ghosts.
- **Flashlight Toggle (Q):** Allows the player to illuminate dark areas and combat the demon.
- **Pause/Menu (Esc):** Allows the player to pause the game, displaying the Pause Menu.

A key rebinding system is not included due to the small and intuitive controls set, along with no requests for this system following testing feedback.

UI Flow

To maximize immersion and limit unnecessary information clutter, the HUD displays only four elements at any given time. The Inventory, Investigation, and Information UI within the Pause Menu provide detailed and immersive interfaces for exploring collectibles in more detail, while the Main Menu and Pause Menu features clear and intuitive layouts for a smooth experience. All menus and HUD elements share a consistent colour palette and theme to ensure visual synchronization. Menus support full mouse navigation for ease-of-use.

Technical Specifications

Engine & Tools

The Phantom Diaries: The Spectral Train was developed with Unreal Engine 5.4, with C++ for core system scripting and Blueprints for supporting features. Rider (coding software), GIMP (image editing software), and Audacity (audio manipulation software) was also used in the creation of the game.

Platform Requirements

	Minimum	Recommended
OS	Windows 10 64-bit	Windows 10/11 64-bit
Processor	AMD Ryzen 5 2600 / Intel Core i5-9400F	AMD Ryzen 5 5600 / Intel Core i5-12400
Memory	8 GB RAM	16 GB RAM
Graphics	NVIDIA GeForce GTX 1060 / AMD Radeon RX 580	NVIDIA GeForce RTX 2060/ AMD Radeon RX 6600
VRAM	4 GB	6 GB
Storage	5 GB available space	8 GB available space (SSD)
DirectX	Version 12	Version 12

Performance Targets

- **Frame Rate:** 60 FPS at 1080p on Minimum Specs, 120 FPS at 1080p on Recommended Specs
- **Resolution:** 1920x1080, supports multiple resolution options
- **Loading Times:** ~1-2 seconds on SSD; ~5-10 seconds on HDD
- **Quality Presets:** Locked at Epic to ensure consistent presentation. Players may adjust Resolution to improve performance

Development & Testing Hardware

Primary Development Machine

OS	Windows 11 64-bit
Processor	AMD Ryzen 7 7800X3D
Memory	32 GB RAM
Graphics	NVIDIA GeForce RTX 4070 SUPER
VRAM	12 GB
DirectX	Version 12

Quality Assurance Testing

The game underwent many rounds of internal playtesting which focused on a variety of elements, such as multiple varied game mechanics and systems, performance stability, options such as resolution changing, font accessibility, brightness and audio, and general game design. Multiple candidates and PCs were utilized for these tests to configure specific performance metrics, while numerous opinions and feedback was delivered on other aspects.

The game went through a variety of tweaks and updates following playtest feedback to alter targeted areas of gameplay to ensure the best experience was created before release build.

Postmortem

What Went Well

- The game showcases all aspects of the original intended design, truly allowing the project to shine in all the imagined areas (horror elements, deep story, intricate world design, puzzle mechanics, and more).
- Playtesting received positive reviews in relation to the imagined areas, confirming the success of the game design.
- The game has been optimized to a small download size while still delivering a high quality, smooth, and enjoyable user experience.
- Production of the game taught and enhanced many new coding and Unreal Engine development techniques that can be applied to later projects.

What Could Have Been Better

- Different versions of the game (Alpha, Beta, Pre-Release, Gold) could have had testing implemented for an easier bug-fixing and gameplay-changing experience. The game was only tested in the Pre-Release version, meaning almost all systems were implemented, removing the ability to easily remake a mechanic or system depending on feedback. For example, some puzzles were revealed to be harder than others and needed more clarity. If this was tested and found in an Alpha or Beta version, then changes to these puzzles would have been easier to complete.
- Earlier stages of development relied on messy, unkempt notes. Maintaining consistent documentation throughout the entire development process would have allowed for easier production and would have streamlined later design referencing and task management.

Key Lessons Learned

- Creating and maintaining a small scope is crucial for the completion of video games, especially when developed solo. Many ideas had to be cut to keep the scope of the project at a realistic size to ensure development could be completed – and within a respectable time range.
- Extensive and vigorous quality assurance testing across all stages of development is critical for the overall finished product, ensuring all bugs – both minor and major – are discovered and fixed to create a professional final version of the game.
- Ensuring all gameplay systems and mechanics are implemented and working as expected before implementing the world itself is a smart and time saving manoeuvre that helps maintain motivation and allows realistic goals to be set and completed.

- Following an intricately detailed game development plan using notes and professional tools, such as Trello, allows for the creation of a proper and easy-to-follow design system which helps keep track of complete and incomplete tasks.

Future Plans

The Phantom Diaries: The Spectral Train is a self-contained experience with no plans for any downloadable content (DLC) packages or future updates. However, many mechanics and storylines were cut during scope refinement – along with a multitude of new ideas – which may form the basis of a future sequel with expanded gameplay and narrative depth in a bigger, even more detailed world. The main goal of a first project in a series is to complete it, allowing future projects in the series to expand upon the laid foundations to deliver new experiences.

References & Inspirations

Historical

The real-world event *The Wreck of the Old '97* served as a loose inspiration behind the story and a strong inspiration behind the setting of the game, utilizing the same time period, locations, and events as the historical occurrence. Early 20th century railway systems (specifically in the golden age of American railway) also served as huge inspiration for the game's setup.

Games

Popular horror game franchises that deliver story-driven, intense gameplay using realistic art styles such as *Outlast* and *Resident Evil* served as strong inspiration for the world building, art direction, and horror elements of the game in an attempt to create the scariest experience possible.

Other

The general life activity of escape rooms serves as a major inspiration for the game's core gameplay design, where exploration-based puzzles are used for the challenges throughout with a countdown timer in the background. My college project some years ago also serves as the main inspiration, which brought these ideas to light in a very small, less attuned manner that stuck with me for many years until this project was created. The college project launched the 'horror escape room on a haunted train' idea which has now been remade following the advancement of learned skills.